

Qatar-10b

R-0139

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_250712 · created 2026-07-16T07:19:10+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460868.896 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.172 +/- 0.016
Transit depth (Rp/Rs)^2	2.96 +/- 0.54 [%]
Orbital Inclination (inc)	80.87 +/- 3.19
Airmass coefficient 1 (a1)	1.113 +/- 0.023
Airmass coefficient 2 (a2)	-0.081 +/- 0.039
Scatter in the residuals of the lightcurve fit is	2.06 %
Best Comparison Star	None
Optimal Aperture	2.51
Optimal Annulus	11.61
Transit Duration (day)	0.086 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.172 ± 0.016 vs 0.1265 (deviation 2.84σ)
Duration fit vs published	0.086 d vs 0.1154 d (deviation 0.89σ)
Mid-time O–C	-23.8 min
Depth SNR	10.7
Residual scatter	2.06 %

Light curve

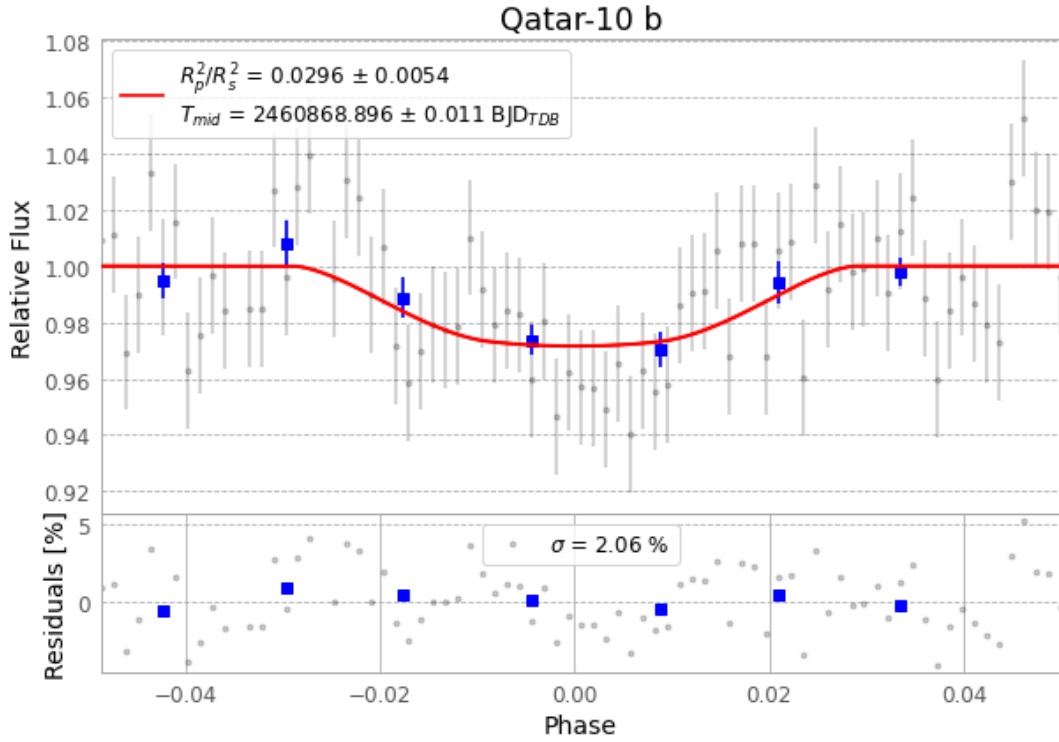


Figure 1 — FinalLightCurve_Qatar-10 b_2025-07-12.png (SHA-256 0b256f214893a48a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 188ca48ad0adfcc1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

79 FITS frames (52 MB), Qatar-10250712073315.FITS → Qatar-10250712112715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-250712.log (c56561a9be6d...), FinalLightCurve_Qatar-10 b_2025-07-12.png (0b256f214893...), FinalLightCurve_Qatar-10 b_2025-07-12.csv (72c1bd04a8cc...)

Compute resources

Wall time 105.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 07:19Z.